Amanat Ur Rahman

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Education

Apr 2027 (expected) — **Ph.D. in Mechanical Engineering** (**Industrial and Systems Engineering** track) — Cincinnati, OH, USA
University of Cincinnati
Dissertation: *Stochastic optimization and decomposition algorithms for large-scale scheduling systems under uncertainty*
Advisor: Dr. Tommaso Giovannelli
Coursework: Optimization; Stochastic Operations Research; Stochastic Optimization Algorithms and Modeling; Industrial Big Data and AI; Lean Six Sigma; Quality Control; Design of Experiments.

Aug 2023 — **M.S. in Industrial Engineering** — Louisville, KY, USA
University of Louisville
Research: *Impact of institutional variability and correlated multi-level factors on post-heart-transplant survivability*
Coursework: Foundations of Optimization I; Algorithms for Optimization; Design of Computer Algorithms; Simulation; Introduction to Machine Learning; Predictive Analytics II; Mathematical Statistics; Engineering Statistics II.

Sep 2020 — **M.S. in Industrial and Production Engineering** — Dhaka, Bangladesh
Bangladesh University of Engineering and Technology (BUET)
Research: *Supply chain performance prediction using grey-based neural network*
Coursework: Probabilistic Methods in Engineering Design; Engineering Data Analysis; Production System Analysis; Advanced Manufacturing Process; Logistics Management.

Dec 2014 — **B.S. in Industrial and Production Engineering** — Dhaka, Bangladesh
Ahsanullah University of Science and Technology (AUST)
Research: *Analysis and suggestions for the supply chain performance of GlaxoSmithKline Bangladesh Limited*
Ranked first among 65 graduating students; Dean's Award.

Research Interests

- **Methodology:** Two-stage stochastic programming, multi-objective and set-valued optimization, decomposition algorithms, mixed-integer linear programming, optimization under uncertainty.
- **Applications:** Healthcare operations management and appointment scheduling, airline network planning and transportation systems, semiconductor manufacturing process control, supply chain analytics.

Publications

Manuscripts in Preparation

[1] **A. U. Rahman**, T. Giovannelli, K. S. Shehadeh, and J. Shi, *Using stochastic programming to mitigate patient no-shows and short-notice cancellations in outpatient clinics through overbooking and walk-ins*, in final co-author revision before journal submission (2026).

[2] **A. U. Rahman** and T. Giovannelli, *Allocating open-access appointment slots in a single-provider session: A two-stage stochastic multi-objective program with an integer first stage*, in preparation (2026).

Refereed Journal Articles

[3] S. M. Ali, **A. U. Rahman**, G. Kabir, and S. K. Paul, *Artificial intelligence approach to predict supply chain performance: Implications for sustainability*, Sustainability 16(6), 2373 (2024).

[4] **A. U. Rahman** and M. T. Zahura, *A grey approach for the prediction of supply chain demand*, American Journal of Industrial Engineering 5(1), 25–30 (2018).

Refereed Conference Proceedings

[5] **A. U. Rahman**, X. Han, and X. Jia, *Enhanced data-driven virtual metrology on chemical mechanical planarization process using dual linear Kalman filter*, Proceedings of the ASME 2024 19th International Manufacturing Science and Engineering Conference (MSEC), Knoxville, TN, USA (2024).

[6] M. R. Bhuiyan, S. Begum, M. A. Sikder, B. M. T. U. Islam, M. Salehin, and **A. U. Rahman**, *An enhanced hybrid grey-neural network model for demand forecasting*, Proceedings of the International Conference on Industrial Engineering and Operations Management, Tokyo, Japan (2024).

[7] **A. U. Rahman**, M. T. Zahura, and A. Rezwan, *Simplified design and fabrication of water sprinkler system: A survey based analysis*, Procedia Engineering 90, 692–697 (2014).

Scholarly Presentations

Conference Presentations

[1] **A. U. Rahman**, T. Giovannelli, K. S. Shehadeh, and J. Shi. Using stochastic programming to mitigate patient no-shows and short-notice cancellations in outpatient clinics through overbooking and walk-ins. 2026 INFORMS Annual Meeting, San Francisco, CA, USA, November 2026 (forthcoming). *(presented by A. U. Rahman)*

[2] **A. U. Rahman**, T. Giovannelli, K. S. Shehadeh, and J. Shi. Using stochastic programming to mitigate patient no-shows and short-notice cancellations in outpatient clinics through overbooking and walk-ins. Modeling and Optimization: Theory and Applications (MOPTA) 2026, Department of Industrial and Systems Engineering, Lehigh University, Bethlehem, PA, USA, August 2026. *(presented by T. Giovannelli)*

[3] **A. U. Rahman**, T. Giovannelli, K. S. Shehadeh, and J. Shi. Using stochastic programming to mitigate patient no-shows and short-notice cancellations in outpatient clinics through overbooking and walk-ins. IISE Annual Conference & Expo 2026, Arlington, TX, USA, May 2026. *(presented by T. Giovannelli)*

[4] **A. U. Rahman**, X. Han, and X. Jia. Enhanced data-driven virtual metrology on chemical mechanical planarization process using dual linear Kalman filter. ASME 2024 19th International

Manufacturing Science and Engineering Conference (MSEC), Knoxville, TN, USA, June 2024. *(presented by X. Jia)*

[5] **A. U. Rahman**, X. Chen, M. Gentili, and J. Trivedi. Impact of institutional variability and correlated multi-level factors on post heart transplant survivability. 2022 INFORMS Annual Meeting, Indianapolis, IN, USA, October 2022. *(presented by A. U. Rahman)*

Poster Presentations

[6] **A. U. Rahman**, X. Chen, M. Gentili, and J. Trivedi. Quantification of contributing factors affecting post heart transplant survivability using multitask Cox model. Research!Louisville, University of Louisville, Louisville, KY, USA, September 2022. *(presented by A. U. Rahman; 3rd place)*

Departmental Seminars

[7] **A. U. Rahman**. Quantification of contributing factors affecting post heart transplant survivability using multitask Cox model. Graduate Student Research Competition, Department of Industrial Engineering, University of Louisville, Louisville, KY, USA, Fall 2022 (2nd place).

Research Experience

Aug 2023 – Present **Graduate Research Assistant** Cincinnati, OH, USA
Department of Mechanical, Materials, and Industrial Engineering, University of Cincinnati

- Developing two-stage stochastic programming models for outpatient appointment scheduling with overbooking, open access, and walk-in patients; solved with an integer L-shaped decomposition algorithm implemented in Gurobi/Python.
- Extending the formulation to a multi-objective two-stage stochastic model within a stochastic set-optimization framework to characterize the Pareto frontier across competing operational objectives.
- Project: Developed a dynamic virtual metrology model based on a Dual Linear Kalman Filter (DLKF) for material removal rate prediction in chemical mechanical planarization, attaining the lowest MSE (7.98) against MLR, EWMA, and Bayesian ARX benchmarks.

Aug 2021 – Jul 2023 **Graduate Research Assistant** Louisville, KY, USA
Logistics and Distribution Institute (LoDI), University of Louisville

- Developed a multi-task Cox proportional hazards model to quantify transplantation center-level factors affecting one-year post-heart-transplant survivability, treating each center as a separate learning task over a shared representation.
- Integrated UNOS donor/recipient records with county- and tract-level area deprivation data (population density, poverty index, stroke mortality) within a 200-mile radius of each center; benchmarked against CoxPH and non-linear CoxPH models by concordance index.
- Authored five online, self-paced instructional modules (IT Basics, Network Foundations, Network Security, Risk Analysis, and Capstone) for a federally funded cybersecurity workforce training program for the logistics industry.

Industry Experience

Jun 2026 – Aug 2026 **Data Scientist Intern (Network Planning)** Erlanger, KY, USA
Atlas Air, Inc.

- Developed a stochastic evaluation framework with MILP recourse (Pyomo, HiGHS, Gurobi) for cargo network planning under operational uncertainty, assessing fleet flow, capacity utilization, and routing.
- Quantified network disruption cost through scenario-based (Monte Carlo) evaluation of the recourse value function.
- Calibrated stochastic parameters from operational data on Azure Databricks and represented the airline route network on Amazon Neptune for connectivity and reachability queries feeding the optimization model.

Jun 2024 – May 2025 **Graduate Research Assistant** Cincinnati, OH, USA
Procter & Gamble (P&G) Digital Accelerator, University of Cincinnati

- Built scalable Azure Databricks pipelines (Python, SQL) translating SAP planning volumes into production-line recipe rates for Personal Care products, reducing annual manual effort by approximately 550 hours (50%).
- Automated data integration from Databricks to Tableau through KNIME and deployed dashboards tracking key performance indicators.

Teaching Experience

Prepared to teach: deterministic and probabilistic operations research, optimization and mathematical programming, stochastic models, discrete-event simulation, engineering economy, production planning and control, supply chain management, engineering statistics and design of experiments.

Apr 2026 **Recitation Instructor** Cincinnati, OH, USA
ISE 3011: Deterministic Systems Models, University of Cincinnati

Jan 2022 – Apr 2022 **Teaching Assistant / Recitation Instructor** Louisville, KY, USA
IE 370: Engineering Economic Analysis, University of Louisville

Jun 2015 – Jul 2021 **Assistant Professor** (Dec 2018 – Jul 2021) Dhaka, Bangladesh
Lecturer (Jun 2015 – Nov 2018)
Industrial and Production Engineering (IPE) discipline, Department of Mechanical and Production Engineering, Ahsanullah University of Science and Technology

Six years of full-time faculty appointment carrying a complete teaching load, centered on the operations research, production systems, and quantitative methods core of the IPE curriculum, with additional laboratory instruction and service teaching to cohorts from other departments. Taught courses fully online during the COVID-19 closure (2020 – 2021).

Operations research and production systems

- **IPE 3107: Operations Research.** Linear programming and the simplex method, duality and sensitivity analysis, transportation and assignment models, game theory, integer and dynamic programming, queuing models.
- **IPE 3205: Production Planning and Control.** Forecasting techniques, purchase and production inventory models, scheduling with Gantt charts,

line balancing, plant location and layout, break-even analysis, time and motion study.

- **IPE 4105: Supply Chain Management.** Sourcing and procurement, distribution strategies, warehouse location and customer allocation, strategic alliances and third-party logistics, the SCOR framework, performance measures.
- **IPE 4103: Operations and Project Management.** Forecasting, capacity planning, inventory models, aggregate planning, MRP and ERP, just-in-time production, line balancing and scheduling, facility location and layout, CPM and PERT.
- **IPE 4206: Industrial Simulation** (laboratory). Discrete-event simulation modeling, input distribution fitting, verification and validation, output analysis, scenario comparison.

Quantitative methods and engineering economics

- **MATH 2207: Numerical Analysis and Statistics.** Root finding, numerical integration and differentiation, curve fitting and interpolation, probability distributions, estimation, hypothesis testing, regression.
- **HUM 2107: Engineering Economy.** Time value of money, equivalence, present and annual worth analysis, rate of return, replacement and depreciation analysis, benefit-cost evaluation.

Design, laboratory, and introductory instruction **IPE 3104: Product Design and Development** (laboratory) and **ME 1103: Introduction to Mechanical and Industrial Engineering**, together with four manufacturing and machine-shop laboratory courses, three of which used laboratory manuals I authored (see *Curriculum Development and Instructional Materials*).

Service teaching for other departments **ME 403: Production Planning and Control** and **HUM 303: Industrial Management**, delivered to Textile Technology cohorts; **HUM 3209: Industrial Management**, delivered to Electrical and Electronic Engineering cohorts. These service courses carried core industrial engineering content (forecasting, capacity and production planning, inventory control, work study, break-even and investment analysis, and organizational behavior) to students outside the IPE major.

Curriculum Development and Instructional Materials

2015 – 2021 **Laboratory manual author** Dhaka, Bangladesh

Department of Mechanical and Production Engineering, Ahsanullah University of Science and Technology

- Authored the laboratory manuals for IPE 4206 (Industrial Simulation), IPE 2102 (Manufacturing Processes), IPE 1210 and IPE 1110 (Machine Shop Practice II and I). All four were adopted as the standing course materials for those laboratories and remained in use across subsequent cohorts.
- Contributed to curriculum revision for the Industrial and Production Engineering program through outcome assessment and course mapping, in support of program accreditation.

Mentoring and Student Supervision

2015 – 2021 **Undergraduate Thesis Supervisor** Dhaka, Bangladesh
Department of Mechanical and Production Engineering, Ahsanullah University of Science and Technology

- Supervised five final-year project and thesis groups from problem formulation through data collection, analysis, and defense.
- Advised students on research methodology, technical writing, and the preparation of results for departmental defense.

Awards and Honors

2026 **Research Travel Grant**, Graduate Student Government, University of Cincinnati, for the 2026 INFORMS Annual Meeting.

2022 **Third place, poster competition**, Research!Louisville, University of Louisville, Louisville, KY, USA.

2022 **Runner-up, Graduate Student Research Competition Award**, Fall 2022 research competition, Department of Industrial Engineering, University of Louisville, Louisville, KY, USA.

2014 **Dean's Award**, Ahsanullah University of Science and Technology, Dhaka, Bangladesh.

2014 **First position in graduating class**, ranked first among 65 graduating students, B.S. in Industrial and Production Engineering, Ahsanullah University of Science and Technology, Dhaka, Bangladesh.

Technical Competencies

Methods: Two-stage stochastic programming; decomposition algorithms; mixed-integer linear programming; multi-objective and set-valued optimization; Monte Carlo scenario generation; survival analysis; state-space filtering; design of experiments

Optimization: Gurobi, Pyomo, HiGHS, Simio (discrete-event simulation)

Programming: Python, PySpark, SQL, Git, LaTeX

Platforms: Azure Databricks, AWS, Amazon Neptune, KNIME, Tableau, SAP, ArcGIS, Excel

Professional Service

2023 – Present **Journal reviewer**

- Operations Research Forum (Springer)
- IEEE Transactions on Industrial Informatics
- Journal of Process Control

2015 – 2021 Contributor to program accreditation activities for the Industrial and Production Engineering discipline, Ahsanullah University of Science and Technology.

Professional Affiliations

Institute for Operations Research and the Management Sciences (INFORMS) 2022–2023, 2026–2027